Midterm

Phil 305 — Propositional and Modal Logic

Part 1: S5

Use the simplified rules for **S5** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **S5**.

1. $\Box(\Box p \to q) \lor \Box(\Box q \to p)$
2. $\Diamond\Box p \to \Box\Diamond p$
3. $(\neg p \lor \neg\Box\Box p) \to \neg\Box p$

Part 2: K

Use the rules for **K** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **K**.

4. $(\Diamond p \land \Diamond q) \to \Box(p \lor q)$
5. $\Diamond(p \to p) \lor \Diamond\neg(q \to q)$
6. $(p \land \neg\neg\Box q) \to (\neg\neg p \land \Box q)$

Part 3: S4

Use the rules for **S4** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **S4**. Remember **S4** is the logic **KT4**, so it has the rules for **T** and **4** as well as **K**

7. $\Box(p \to q) \to (\Diamond\Diamond p \to \Diamond q)$
8. $(p \land \Diamond\Box p) \to \Box p$

Name: ....

1. $\Box(\Box p \to q) \lor \Box(\Box q \to p)$ **in S5**

2. $\Diamond\Box p \to \Box\Diamond p$ in S5

3. $(\neg p \vee \neg\Box\Box p) \rightarrow \neg\Box p$ **in S5**

4. $(\Diamond p \wedge \Diamond q) \rightarrow \Box(p \vee q)$ **in K**

5. $\Diamond(p \to p) \lor \Diamond\neg(q \to q)$ **in K**

6. $(p \wedge \neg\neg\Box q) \to (\neg\neg p \wedge \Box q)$ **in K**

7. $\Box(p \to q) \to (\Diamond\Diamond p \to \Diamond q)$ in S4

8. $(p \wedge \Diamond\Box p) \to \Box p$ in S4